ROS Community

Open Source Ecosystem



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Agenda

- Open Source Robotics Alliance (OSRA)
- ROS Community / Governance
- ROSCon and Regional ROSCons (ROSCon JP)
- Sponsorship / Participation
- ROS Japan User Group
- ROS Japan Developer Kaigi
- Open Source Ecosystem: What's in it for us?

History of ROS

Year	Milestone
2012	**Open Source Robotics Foundation (OSRF)** incorporated; first **ROSCon** (St. Paul, USA)
2017	OSRF + OSRC become **Open Robotics**; first ROS 2 release **Ardent Apalone**
2020	ROS 1 final release, **Noetic Ninjemys**
2022	**Intrinsic** (Alphabet) acquires OSRC, not OSRF, not ROS
2024	**OSRA** launched to govern ROS, Gazebo, Open-RMF and Infrastructure
2025	ROS 1 Noetic reaches End of Life
2026	**Lyrical Luth** released, LTS until May 2031

ROS 1 to ROS 2: Why the Rewrite?

- ROS 1 was built for a single research robot in a lab network.
- Industry needed what ROS 1 could not give:
  - Real-time capable, embedded and microcontroller support
  - Security (authentication, encryption, access control)
  - Multi-robot, lossy and multi-platform networking
  - Standards-based middleware (DDS), pluggable RMW
  - Production quality: lifecycle nodes, QoS, Windows / macOS / RHEL
- ROS 2 is designed **with** industry, which is exactly why industry needs to keep participating.

ROS Today in Numbers

From the 2025 ROS Metrics Report:

- **~984 million** package downloads in 2025, **+85%** over 2024
- **1.3 million** unique IPs downloading packages in October 2025, up from 0.84 million one year earlier
- **91%** of downloads are ROS 2
- ROSCon 2025 Singapore: **950+ attendees, 51+ countries, 354+ companies**

ROS is the de facto common language of robotics software.

Active ROS 2 Distributions

Distribution	Release	End of Life	Note
Rolling Ridley	Continuous	N.A	Development mainline
Lyrical Luth	May 2026	May 2031	Latest **LTS**
Kilted Kaiju	May 2025	Nov 2026	Non-LTS
Jazzy Jalisco	May 2024	May 2029	LTS
Humble Hawksbill	May 2022	May 2027	LTS

- New release every year on **World Turtle Day (May 23)**, LTS every two years.
- A five-year LTS only happens because someone funds and maintains the build farm, backports and security fixes.

Open Source Robotics Alliance

- An initiative of the **OSRF**, announced 18 March 2024, operating since 15 April 2024
- OSRF assigned governance of its projects to OSRA: **ROS, Gazebo, Open-RMF, Infrastructure, ros-controls**
- **Mixed membership + meritocratic** model, following Linux Foundation / Eclipse / Apache style foundations
- Goal: *long-term stability and health* of the projects, funded by members, governed with the community



ROS Community / Governance

- **Meritocracy**: influence comes from sustained contribution, not from the size of your company.
- **Committers** are nominated and mentored, then accepted by the PMC.
- **PMC Members** vote on project matters; by default every committer is a PMC member.
- **Project Leader** is nominated by the PMC and accepted by the TGC, re-elected yearly.
- Big changes go through **enhancement proposals** (REPs / design docs) and public review.
- Minutes of formal decisions are published.

Where the Community Lives

- GitHub ros2: source, issues, pull requests, design
- Open Robotics Discourse: announcements, releases, discussions, **local user groups**
- Robotics Stack Exchange: Q&A
- Open Robotics Zulip: real-time chat, working groups, Japan channel
- ROS Index and docs.ros.org: packages and documentation
- Working Groups and SIGs: Client Libraries, rosgraph, Release, Security, Interoperability, Embedded, Navigation, Control, ...

ROSCon

Year	City	Year	City
2012	St. Paul, USA	2019	Macau
2013	Stuttgart, Germany	2020 / 2021	ROS World (online)
2014	Chicago, USA	2022	**Kyoto, Japan**
2015	Hamburg, Germany	2023	New Orleans, USA
2016	Seoul, Korea	2024	Odense, Denmark
2017	Vancouver, Canada	2025	Singapore
2018	Madrid, Spain	2026	Toronto, Canada

ROSCon 2025 Singapore

- 950+ attendees, 51+ countries
- 354+ companies, 66+ universities
- **56 sponsors**, 10 sponsored mini-workshops
- 464 workshop attendees, 8 BoF sessions



ROSCon 2026 Toronto

- 22 – 24 September 2026, Sheraton Centre Toronto

Regional ROSCons: Local Language, Local Network

2026 Event	When	Where
ROSCon FR	22 – 25 June	Paris, France
ROSCon JP	**4 – 5 August**	**Tsukuba, Japan**
ROSCon China	16 – 17 October	Shanghai, China
ROSCon UK	21 – 23 October	Edinburgh, UK
ROSCon SG	26 October	Singapore
ROSCon BE	25 – 26 November	Nivelles, Belgium
ROSCon Türkiye	3 – 4 December	Istanbul, Türkiye

ROSCon JP

Year	Venue	Year	Venue
2018	Tokyo (1st)	2023	Tokyo
2019	Tokyo	2024	Tokyo
2021	Online	2025	Nagoya (Port Messe)
2022	Kyoto	2026	Tsukuba, **2 days**

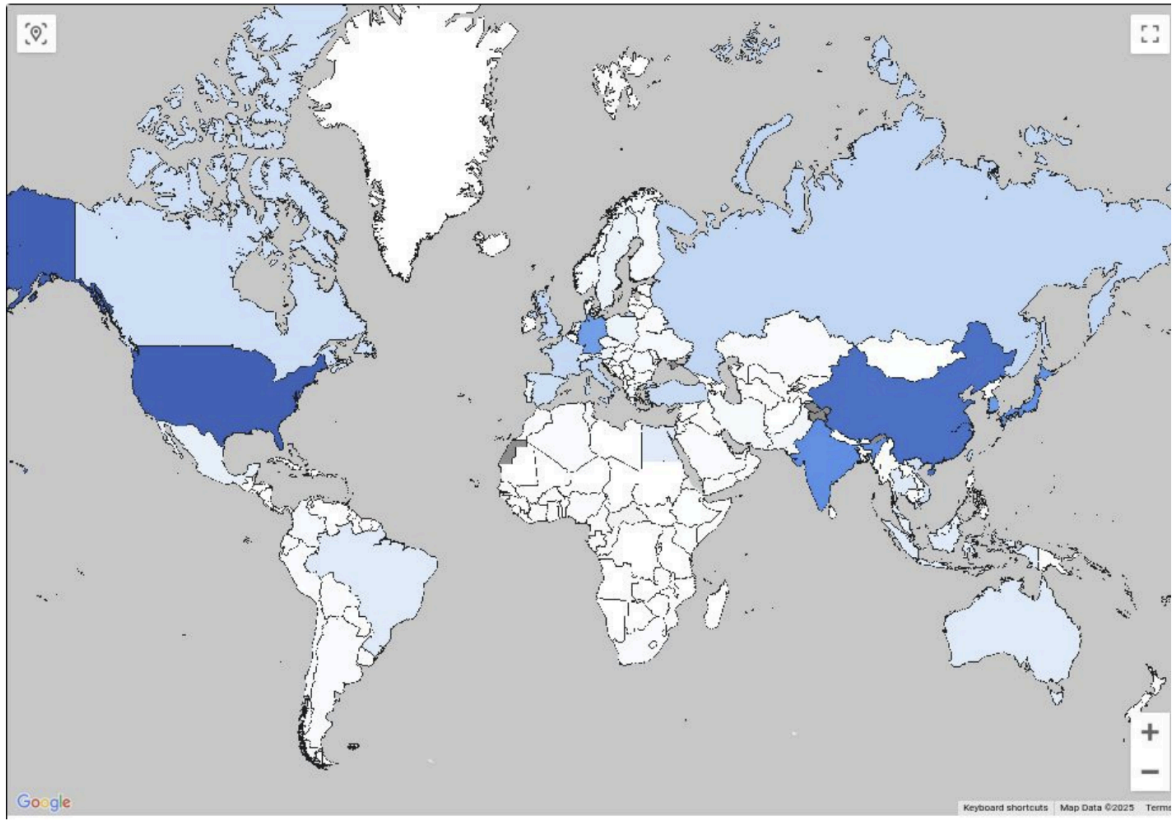
Sponsorship / Participation

1. **Attend** ROSCon JP / ROSCon, send your engineers
2. **Speak**: submit a talk or lightning talk about what you built
3. **Sponsor** ROSCon JP or ROSCon
4. **Join OSRA** as Supporting Organization, Silver or higher
5. **Contribute** code, docs, issues, reviews upstream
6. **Host** a ROS Japan UG meetup at your office
7. **Let your engineers join** Developer Kaigi and working groups on company time

ROS Japan User Group

- Volunteer-run community for ROS users in Japan, since **2016**
- **3,390 members** on connpass, **73 events** held (Nov 2026)
- Study sessions, lightning talk nights, hackathons, in-person meetups (Tokyo, Osaka, ...)
- Recent: #64 3D Gaussian Splatting (Sep 2026), #63 LT night (Jul 2026), #62 Vibe Coding hackathon (May 2026)
- Organizes **ROSCon JP**
- Home on Discourse: Local User Groups / Japan





wiki.ros.org

Traffic volume by country

	Country	Active users	New users
	Total	1,348,690 53.29% of total	982,138 41.08% of total
1	China	235,150	163,198
2	United States	217,801	153,364
3	India	100,771	72,687
4	Japan	99,109	65,420
5	South Korea	90,264	64,628
6	Germany	75,428	50,077
7	Hong Kong	43,374	23,260
8	Taiwan	34,939	22,488
9	United Kingdom	34,443	24,373
10	Singapore	33,967	20,051
11	Russia	31,598	23,024
12	France	27,734	18,418
13	Italy	27,030	18,617
14	Türkiye	26,473	19,117
15	Canada	26,166	18,324
16	Spain	21,160	14,257
17	Vietnam	17,878	12,885
18	Australia	16,712	11,525
19	Thailand	16,349	11,966
20	Brazil	15,229	10,436

ROS™

ROS Japan Developer Kaigi

ROS本線を開発しよう 🤖 技術課題をシェアしよう 👷 目指せ！ROSコミッター＆メンテナー 🚀

- Japanese-language, **online, bi-weekly, 50 minutes**, no recording
- Scope: **ROS 2 core repositories** (rclcpp, rclpy, rcl, rmw, ros2cli, rosbag2, docs, ...)
- First meeting: **16 September 2025**
- In-person edition at University of Tsukuba, **3 August 2026**, the day before ROSCon JP

Open Source Ecosystem: What's in It for Us?

- **Shared, non-differentiating infrastructure**: middleware, build tools, drivers, navigation, control, simulation
- Hundreds of companies split the cost of what none of them sells
- Each company differentiates **above** the shared layer: the application, the hardware, the integration
- The shared layer only stays healthy if enough of the beneficiaries give back time or money

Concrete Returns for a Company

- **Lower cost**: fixes upstream once, instead of a private fork
- **Lower risk**: LTS schedule, security policy and EOL
- **Interoperability**: sensors, arms, bases and tools from hundreds of vendors already speak ROS 2
- **Talent**: engineers, ROS skills are portable, so hiring is easier
- **Influence**: a committer or a member vote shapes the roadmap toward your use cases
- **Reputation**: customers and partners see you as a contributor, which matters in procurement and partnerships

Summary

- ROS has outlived three stewards because it belongs to its community; **OSRA** now gives that community formal governance and funding.
- **ROSCon** and regional **ROSCon JP** are where the roadmap, the people and the hiring happen.
- Japan is one of the biggest ROS user bases, with very little presence upstream. **ROS Japan UG** and **ROS Japan Developer Kaigi** are the on-ramps.
- Participation and sponsorship are inexpensive compared with the cost of passively depending on a project you do not influence.

Let's build ROS, not only build on ROS. 🚀

References

- ROS 2 Documentation / Governance / Releases
- Open Source Robotics Alliance / Membership / Policies and Procedures
- Announcing the OSRA (Mar 2024)
- 2025 ROS Metrics Report
- ROSCon / ROSCon JP
- ROS Japan User Group (connpass) / Discourse Japan
- ROS Japan Developer Kaigi
- Why ROS 2? / A Decade of Open Robotics